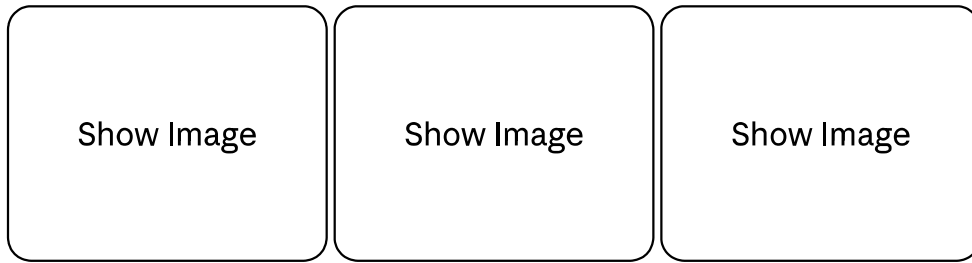


Data Cleaning & Visualization Dashboard

An interactive React dashboard that demonstrates a full **data cleaning pipeline** followed by rich visualizations — built with Recharts and styled with a dark theme.



Features

-  **6-step cleaning pipeline** — removes duplicates, fixes outliers, imputes missing values
-  **5 visualization tabs** — overview, salary analysis, dept breakdown, correlation scatter, raw table
-  **Dark UI** — custom dark theme with department color coding
-  **Fully client-side** — no backend needed, all logic runs in the browser

Project Structure

```
src/  
├─ data.js           # Raw (dirty) dataset & color constants  
├─ cleaning.js       # cleanData() pipeline + stats() helper  
├─ components.jsx    # Reusable UI: StatCard, CustomTip, TabBar  
└─ App.jsx           # Main app with all tab views
```

Cleaning Pipeline

Step	Action	Method
1	Remove duplicates	Key: <code>name + dept + joined</code>
2	Fix age outliers	Ages > 100 or < 16 → <code>null</code>

Step	Action	Method
3	Fix negative salaries	Negative values → <code>null</code>
4	Impute missing age	Replace with column mean
5	Impute missing salary	Replace with department median
6	Impute missing score	Replace with department mean

Dashboard Tabs

Tab	Description
Overview	KPI cards, missing value bars, headcount pie chart
Cleaning Log	Step-by-step pipeline results with record counts
Salary Analysis	Distribution histogram + avg salary by department
By Department	Avg score bar chart + per-dept summary cards
Correlation	Age vs. Salary scatter plot colored by department
Clean Data	Full cleaned dataset in a sortable table

Getting Started

Prerequisites

- Node.js ≥ 16
- npm or yarn

Installation

```
bash
```

```
# Clone the repository
git clone https://github.com/your-username/data-cleaning-dashboard.git
cd data-cleaning-dashboard

# Install dependencies
npm install

# Start the dev server
npm run dev
```

Dependencies

```
bash

npm install recharts
```

Built With

- [React](#) — UI framework
- [Recharts](#) — chart library
- [Google Fonts](#) — Inter, DM Mono, Syne

Dataset

The dataset (`data.js`) contains **20 intentionally dirty employee records** with:

- `null` values in `age`, `salary`, and `score`
- An age outlier (`999`)
- A negative salary (`-5000`)
- Duplicate entries

After cleaning, **18 valid records** remain across 4 departments: Engineering, Marketing, HR, and Management.

License

This project is licensed under the [MIT License](#).

Contributing

Pull requests are welcome! For major changes, please open an issue first to discuss what you'd like to change.